QC STORY PROJECT

PROJECT TITLE :- To Eliminate Knifing Of C Pillar Rhs With Rear Floor Frame Assembly

Mahindra and Mahindra Ltd,_Nashik Plant, Scorpio Body.

	Plan	Actual
Start Date	**11.05.12**	**17.05.12**
Completion Date	**11.08.12**	**21.09.12**

Current Status:

Step-9 Completed

Cross Functional Team

No	Description	Name	Designation	Dept	Phone	Email
1	**Team Leader**	Gokul Amrutkar	DY Manager	MQA	9922408133	Amrutkar.gokul@mahindra.com
2	**Mentor**	Sandeep Deshpande	Manager	MQA	9422770889	Deshpande.sandeep2
3	**Member 1**	Vivek Bhat	DY Manager	Tooling		
4	**Member 2**	Amit Kulkarni	DY Manager	Prod		
5	**Member 3**	Mangesh Gangurde	Asst Manager	MQA		
6	**Member 4**	Giri	Asst Maanger	SQA		

1. Select the Theme

Brief description of concern

What :- “C” pillar area knifing occurs with respect to rear floor ,leads into excess gap between rear wheel house to rear floor

Where:- Ballooning stage (Z22#10)

When:- In any shift (No co-relation with shift)

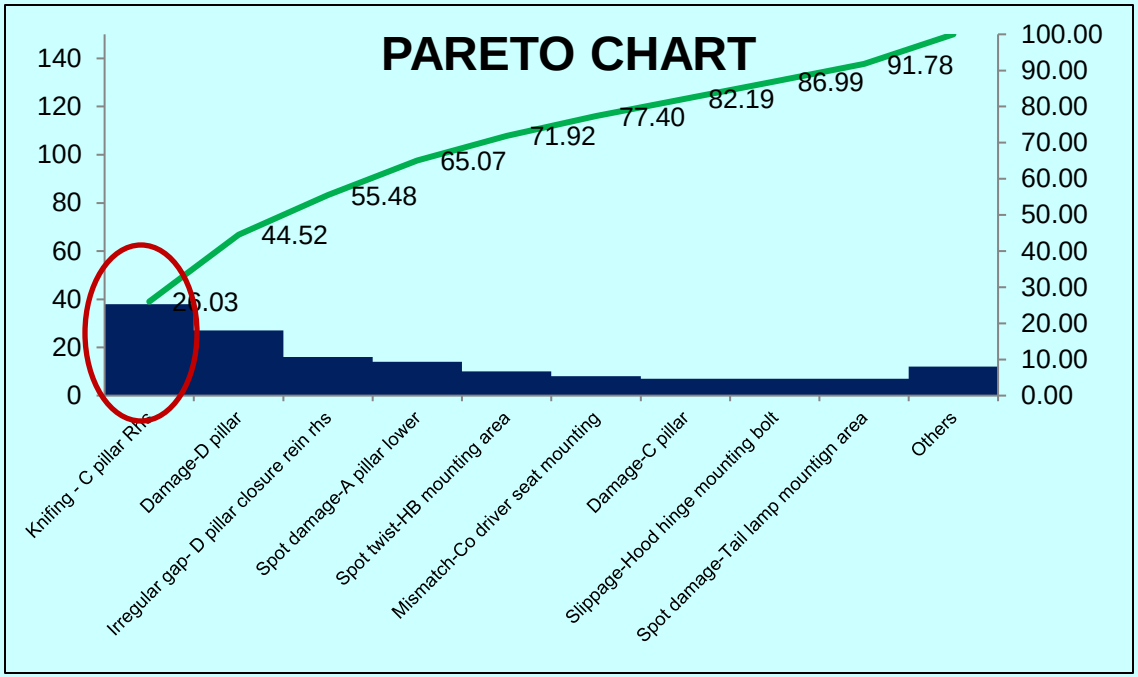
How :- 14 Incidences in April-12

Current level :- 12 Incidences/ Month

Loss of BIW :- 9 in April 2012

2. Justify the choice (reasons)

Emerging concern reported from TCF on dated 30.04.2012

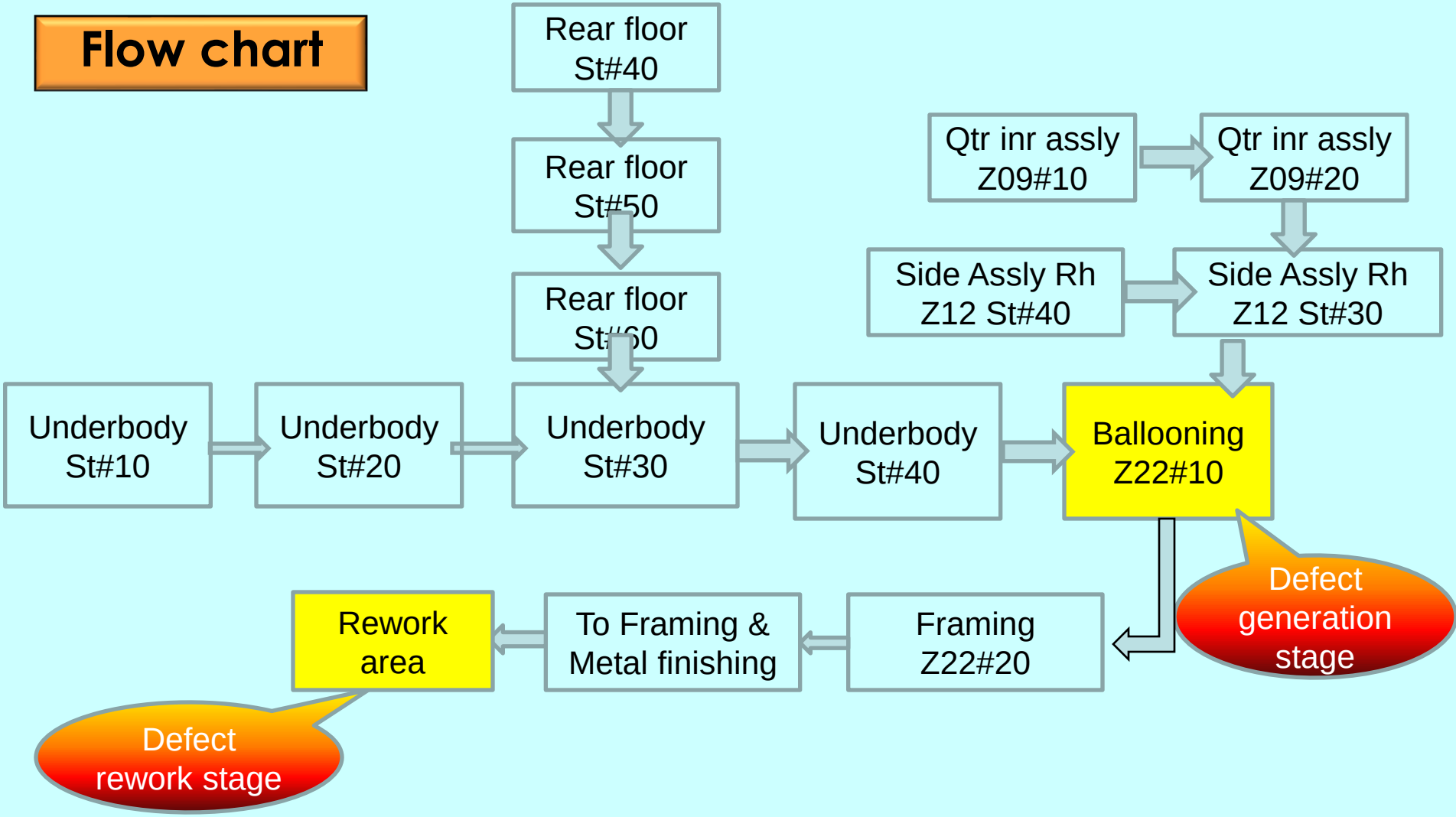


9 BIW loss in April 12 related to Quality concern
Cost is (47000*9=4.23 Lacs)

2.b Immediate Containment Action

- DRF given to ballooning stage operators –If knifing occurs on stage write on traveler card for rework notification

3. Understand the current situation



3. Understand the current situation

Conformity Analysis
(L0 Analysis)

PU/ Area : S B J
Cell : Framing

Requested By : MQA | Date - 30/4/12 | Source - TCF | Conformity Analysis Sr.No CA2P 575

Problem / Concern Statement :
Excess gap Rear floor to W/H RHS.

FMEA Severity / : Gravity : ☐ V1+ ☑ V1 ☐ V2 ☐ V3

No of Incidences on the day (I) : 1

Concern Priority Number : 74.

Vehicle/ Aggre. No. 17464

Date of Incidence : 28/4/12

Shift / Time : I

Whether SC/CC/Z parameter affected ? ☐ YES ☑ NO ☐ Z ☐ CC ☐ SC

Immediate containment action for Customer Protection :
– sweep away done.

Does a related Conformity Analysis or QC story already exist ? ☐ YES ☑ NO | If Yes | Confor | QC St

Failure mode or cause/s identified - –

Problem Assigned to - Umesh Kankal BWT : - Framing.

Stage / Post Analysis -

Does a control already exist ? ☐ YES ☐ NO

Man (Human actions)		Method (Process)		Others	
Is operator trained ?	☑ YES ☐ NO	Main steps are respected ?	☑ YES ☐ NO	Is there a SOS/ SOP / CP/monitoring plan ?	☑ YES ☐ NO
Operator level (I.L.U)	1	Is there a related key point in SOP/CP/SOS ?	☐ YES ☑ NO	Is it implemented ?	☑ YES ☐ NO
OR Dexterity level(1 . . 4)	3	Are the reasons for key point known ?	☐ YES ☑ NO	Does it conform ?	☑ YES ☐ NO
Is operator under training ?	☐ YES ☑ NO	Are operating reqmnts respected ?	☑ YES ☐ NO	Is the packaging in good condition ?	☑ YES ☐ NO
Is operator temporary for Stage ?	☐ YES ☑ NO	Is the defect part of the checkman/ Buy off list ?	☐ YES ☑ NO	Last prevetive measures taken on -	
Bad rework ?	☐ YES ☑ NO	Was it detected by Inspector/ Buoff Opr ?	☐ YES ☑ NO	Is there any alert mechanism for failure ?	☐ YES ☐ NO
Is the specified facility used ?	☑ YES ☐ NO	In only one shift ?	☐ YES ☑ NO	Is the specified product used ?	☐ YES ☐ NO
Last calibration dates		In only one line ?	☐ YES ☑ NO	Old it evolve / Change ?	☐ YES ☐ NO
Facility in good condition ?	☑ YES ☐ NO	The problem follows the vehicle ?	☑ YES ☐ NO	Is the storage SOP respected ?	☐ YES ☐ NO
Is it a substitute facility ?	☐ YES ☑ NO	The problem follows the part ?	☐ YES ☑ NO	Is the shelf life / expiry date respected ?	☐ YES ☐ NO
Are machine parameters known ?	☑ YES ☐ NO	Is it a repeat problem ?	☐ YES ☑ NO		
Parameters are respected ?	☑ YES ☐ NO				
Machine		**Product Failure**		**Material / Consumable**	

Note : 1. If both fields are kept unmarked, it means that mentioned factor is not applicable for this concern)

Analysis Conclusion (Attach Sketch/ Photograph if necessary) -

No Nonconfirmity observed in L0 ananlysis.

3. Understand the current situation

Is Level 1 analysis required ? ☑ YES ☐ NO

Actions -

1.Containment Action	Resp	Date	Status
DRF to be given to all stage operators	Umesh	30/4	Cls

2. Permanent Action -	Resp	Date	Status

Analysed by Name/s - Sandip / Umesh / Diwakar

Date - 30/4/12

Choice : ☐ MSS ☑ QC Story

Format No. MQS/MAPS/F01, Rev 00

3. Understand the current situation

Process Flow

Z08#20 output

Rear floor assembled

Spot welding with rear floor

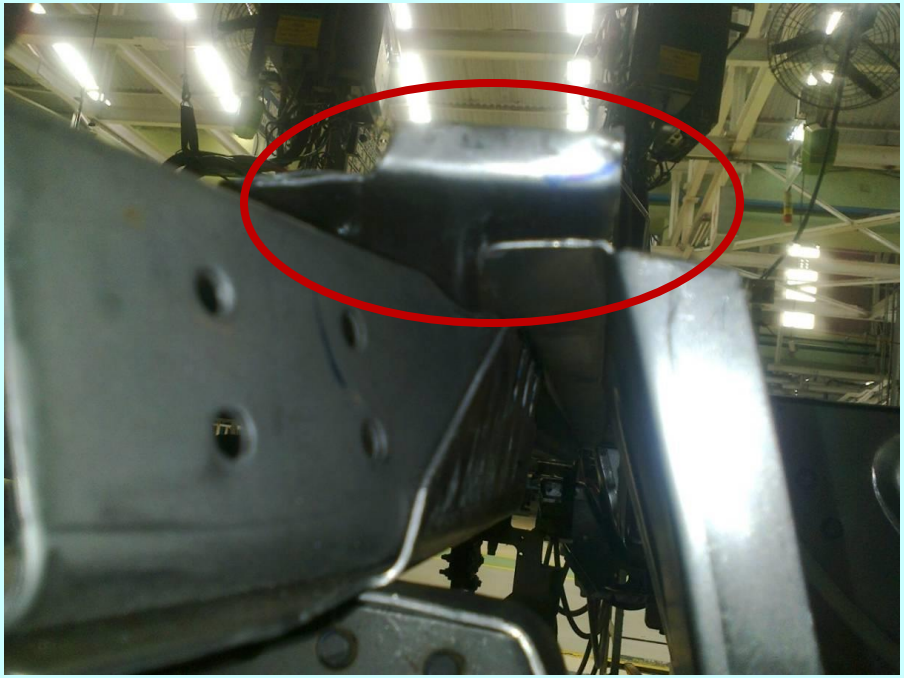
Side assly output Z12#40

Side assly locating in fixture

Wheel hsg profile matching with rear floor

3. Understand the current situation

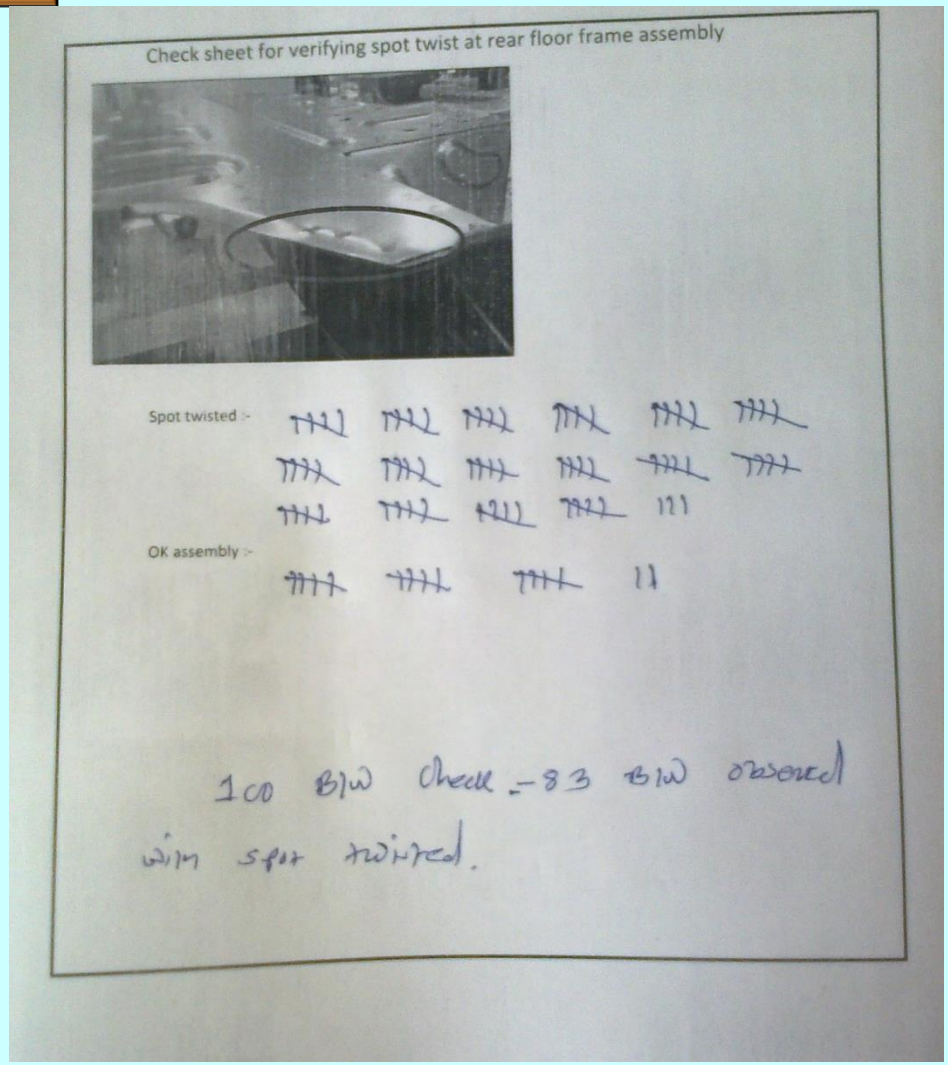
Observation of process



Center floor flange fouling to cross member frame

3. Understand the current situation

Observation of process



Check sheet for verifying spot twist at rear floor frame assembly

Spot twisted :-

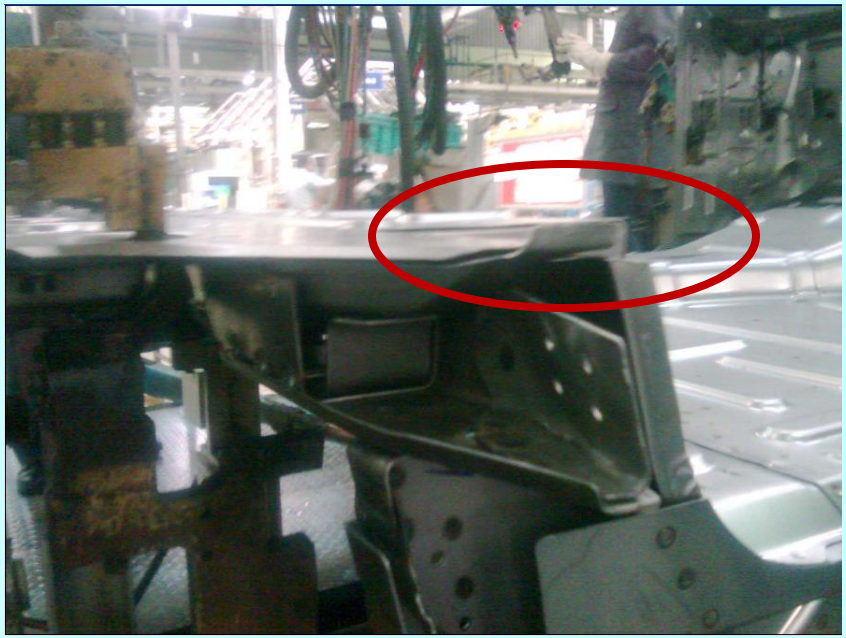
OK assembly :-

100 B/W check - 83 B/W observed with spot twisted.

3. Understand the current situation

Observation of process

Flange damage during spot welding



4. Select Target

Present Level:

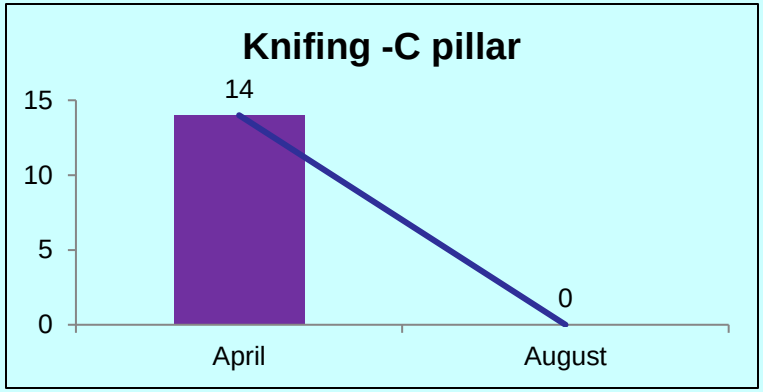
Avg. 14 nos /Monthly knifing concern reported

Avg. 9 BIW's / Monthly loss of BIW (Cost -5.64 lacs)

Target:

FTBO Level : 0 incidences

BIW loss –Zero BIW Loss (Due to C pillar Rhs knifing)



5. Analysis

Brainstorming by the CFT

Sr No	Observation	Possible cause
1	Observations related to parts	1.Rear wheel house profile at mating area
		2.Quarter inner panel assembly dimensions
		3.Side assembly dimensions at locating area
		4.Rear floor frame assembly dimensions
		5.Sill side assembly cap position
		6.Sill side assembly position in fixture

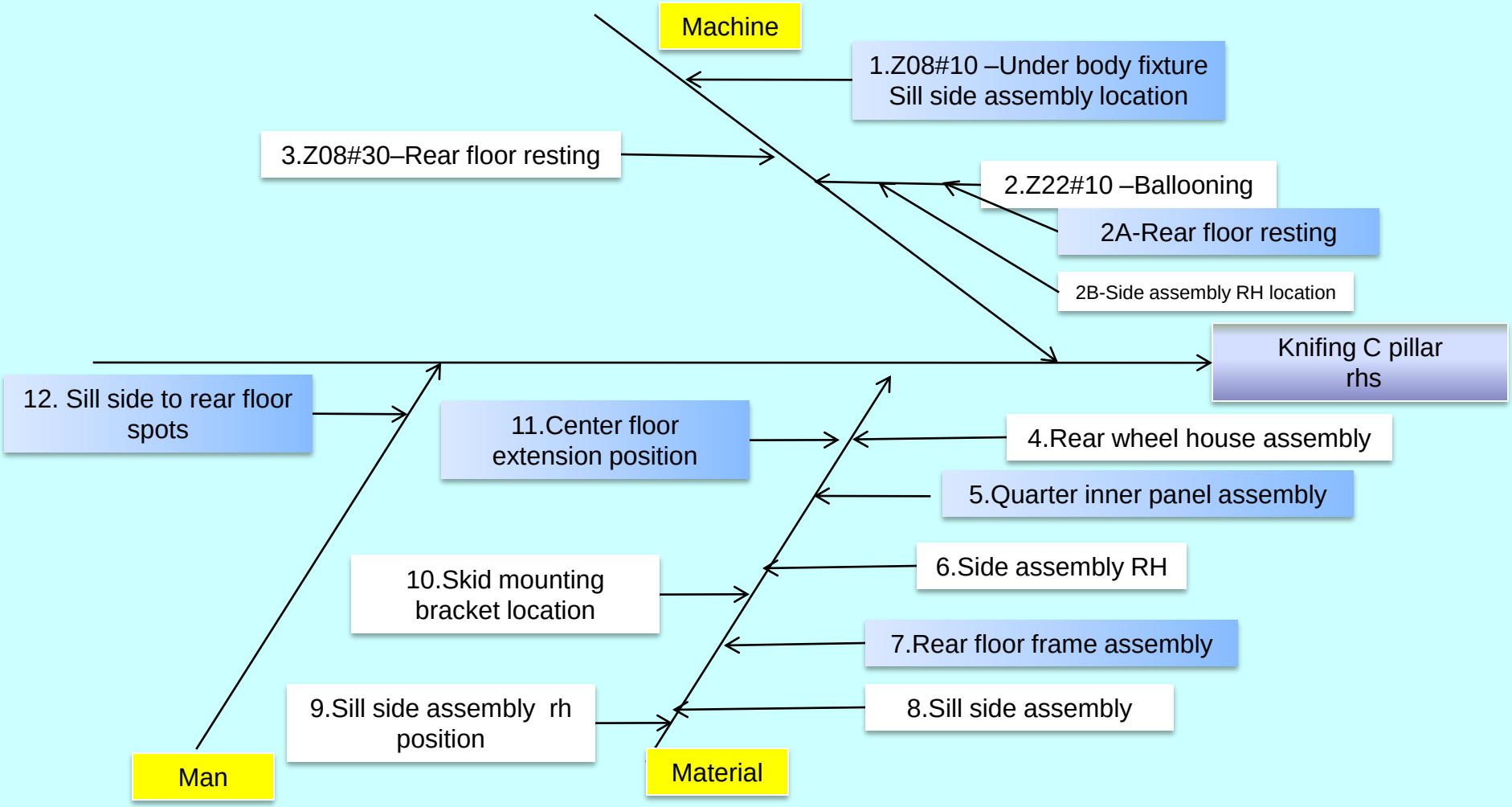
5. Analysis

Brainstorming by the CFT

Sr No	Observation	Possible cause
1	Observations related to parts	7.Center floor extension position
		8.Skid mounting bracket location
2	Fixture co-ordinates	9.Z08#10 –Sill side assembly location
		10.Z22#10 –Rear floor resting
		11.Z22#10 –Side assembly RH location
		12.Z08#30-Rear floor resting
3	Spot welding	13.Z08#30 sill side to rear floor spots surface

5. Analysis

5.1 Identify possible causes:



TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
1	Z08#10 –Under body fixture Sill side assembly location	Confirm the front floor resting & sill side assembly locating points in calibration reports. In Z axis deviation observed 0.1 mm.	**Valid**

C_SILL_SIDE_FRT_LOC_PIN_LH *(Constructed Circle)*

		Measured	Nominal	+Tol	-Tol	Dev	Out of Tol
	X	1856.84	1857	0.5	-0.5	-0.16	*******
Center	Y	-680	-680	0.5	-0.5	0	*******
	Z	568.43	568	0.5	-0.5	0.43	*******

C_SILL_SIDE_RR_LOC_PIN_LH *(Constructed Circle)*

		Measured	Nominal	+Tol	-Tol	Dev	Out of Tol
	X	3115.44	3115	0.5	-0.5	0.44	*******
Center	Y	-680	-680	0.5	-0.5	0	*******
	Z	543.4	544	0.5	-0.5	-0.6	-0.1

C_SILL_SIDE_FRT_LOC_PIN_RH *(Constructed Circle)*

		Measured	Nominal	+Tol	-Tol	Dev	Out of Tol
	X	1857.34	1857	0.5	-0.5	0.34	*******
	Y	680	680	0.5	-0.5	0	*******
Center	Z	568.46	568	0.5	-0.5	0.46	*******

C_SILL_SIDE_RR_LOC_PIN_RH_ *(Constructed Circle)*

		Measured	Nominal	+Tol	-Tol	Dev	Out of Tol
	X	3115.5	3115	0.5	-0.5	0.5	*******
	Y	680	680	0.5	-0.5	0	*******
Center	Z	543.56	544	0.5	-0.5	-0.44	*******

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

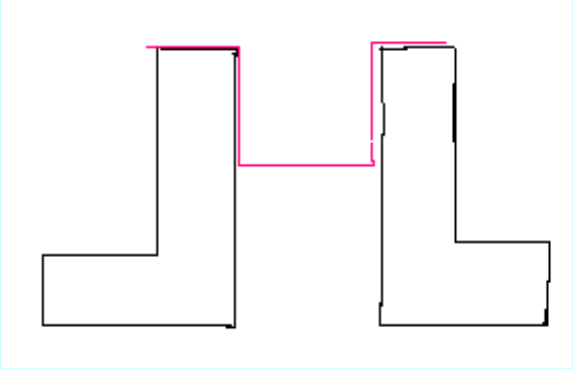
No	Probable Cause	Testing and Observations	Conclusion
2	2.Z22#10 – Ballooning	Cross member locating & body mounting locating found ok.	Invalid
2A	2A-Rear floor resting Z22#10	Rear floor resting miler as compared to LH,RH miler fouling to cross member in X axis	**Valid**
2B	2B-Side assembly RH location	Side assembly location found ok as per Faro arm inspection report	Invalid

M_BODY_MTG_02_PIN_LH *(Measured Circle , number of points 8)*

		Measured	Nominal	+Tol	-Tol	Dev	Out of Tol
	X	1640.25	1640	0.5	-0.5	0.25	*******
	Y	-328	-328	0.5	-0.5	0	*******
Center	Z	-300	-300	0.5	-0.5	0	*******

M_RR_FLR_RR_X-MBR_LOC_PIN-RH *(Measured Circle , number of points 3)*

		Measured	Nominal	+Tol	-Tol	Dev	Out of Tol
	X	4076.52	4076	0.5	-0.5	0.52	0.02
	Y	384.32	384	0.5	-0.5	0.32	*******
Center	Z	-300	-300	0.5	-0.5	0	*******



TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

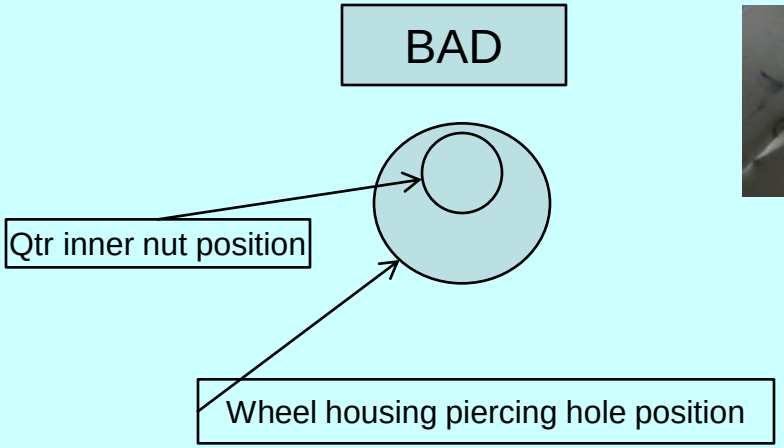
No	Probable Cause	Testing and Observations	Conclusion
3	Z08#30–Rear floor resting	Rear floor resting miler position confirmed in Z axis found ok.	Invalid
4	Rear wheel house assembly	Rear wheel house assembly checked on checking fixture; For location no 5 reading observed 3.7 mm which is beneficial to concern	Invalid

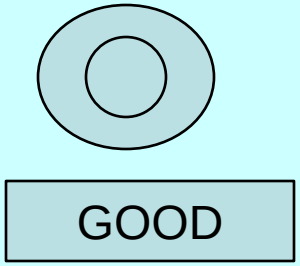
SURFACE GAP AT FLANGE				
POINTS	SPECIFIED- 3+/- 0.5			
	SAMPLE-1	SAMPLE-2	REMARK	
1	2.5	2.5	OK	
2	3	3.1	OK	
3	3.5	3.5	OK	
4	3.6	3.5	OK	
5	3.7	3.7	NOT OK	
6	3.4	3.4	OK	
7	3	3	OK	
8	2.8	2.8	OK	
9	2.6	2.6	OK	
10	2.5	2.6	OK	
11	2.7	2.7	OK	

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
5	Quarter inner panel assembly – nut mismatch	For quarter inner panel assembly good & not good comparison carried with LH quarter panel assembly. As per LH quarter inner panel assembly ,rh side nut position not concentric with wheel house	**Valid**

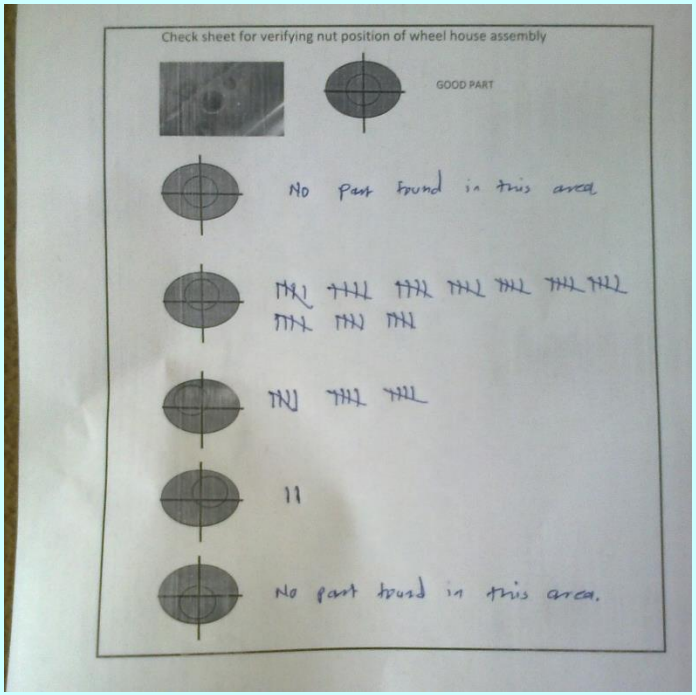




TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
5	Quarter inner panel assembly – nut mismatch	For quarter inner panel assembly good & not good comparison carried with LH quarter panel assembly. As per LH quarter inner panel assembly ,rh side nut position not concentric with wheel house	**Valid**



TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
6	Side assembly RH	Side assembly RH & LH position confirmed in CMM which is found OK	Invalid
7	Rear floor frame assembly	Rear floor frame assembly results confirmed at M/S Musco found OK.	Invalid

DESCRIPTION OF PART		LH/RH	AXIS	NOMINAL	UPR TOL	LWR TOL	17-Apr	11-Apr	01-Apr
							SC 15975	SC15136	SC12616
RING PNL LOC-LH	ALL41	LH	X	1665.00	1.00	-1.00	-0.6	-0.1	-0.2
			Y	-781.80	1.00	-1.00	-0.8	-0.5	-0.7
			Z	890.00	1.00	-1.00	1.0	0.8	1.0
RING PNL LOC-RH	ALR41	RH	X	1665.00	1.00	-1.00	-0.4	0.1	0.2
			Y	781.80	1.00	-1.00	0.4	-0.1	0.5
			Z	890.00	1.00	-1.00	0.4	-0.1	0.2
LOC SLOT ON RING PLN-LH	BLL42	LH	X	3490.00	1.50	-1.50	-0.8	-0.9	-1.1
			Y	-823.90	1.50	-1.50	*2.4*	1.2	0.8
			Z	1092.00	1.50	-1.50	-0.3	-0.9	0.4
LOC SLOT ON RING PLN-RH	BLR42	RH	X	3490.00	1.50	-1.50	-1.3	-0.1	-0.5
			Y	823.90	1.50	-1.50	1.3	0.7	1.4
			Z	1092.00	1.50	-1.50	-0.2	-0.5	0.3

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
8	Sill side assembly	Sill side assembly cap position check –results are found ok.	Invalid

RH	LH
36.85	36.88
36.50	36.90
36.50	36.88
36.90	36.88
36.80	36.83
36.88	36.95
36.60	36.90
36.75	36.88
36.72	36.83
36.90	36.87
36.85	36.90
36.88	36.92
36.90	36.84
36.81	36.85
36.88	36.88

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
8	Sill side assembly	Sill side assembly cap position check –results are found ok.	Invalid

RH	LH
65.6	65.0
66.7	65.8
64.2	65.2
66.7	66.0
65.9	65.5
66.4	65.4
66.6	65.3

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
9	Sill side assembly rh position	Sill side assembly Rh position confirmed on CMM with respect to foot step mounting bracket which is found ok.	Invalid

DESCRIPTION OF PART		LH/RH	AXIS	NOMINAL	UPR TOL	LWR TOL	11-Apr SC15136	01-Apr SC12616
FOOTSTEP MTG FRT ON SILL SIDE-LH		LH	X	1745.00	1.50	-1.50	0.1	-0.4
			Y	-659.60	1.50	-1.50	1.4	-1.5
			Z	532.80	1.50	-1.50	-0.9	0.2
FOOTSTEP MTG FRT ON SILL SIDE-RH		RH	X	1745.00	1.50	-1.50	1.0	0.0
			Y	659.60	1.50	-1.50	-0.5	0.2
			Z	532.80	1.50	-1.50	1.0	-0.9
FOOTSTEP MTG MIDDLE ON SILL SIDE-LH		LH	X	2335.00	1.50	-1.50	0.5	1.3
			Y	-659.60	1.50	-1.50	0.4	0.5
			Z	532.80	1.50	-1.50	1.0	-0.1
FOOTSTEP MTG MIDDLE ON SILL SIDE-RH		RH	X	2335.00	1.50	-1.50	1.1	1.5
			Y	659.60	1.50	-1.50	1.2	-1.5
			Z	532.80	1.50	-1.50	1.3	-1.1
FOOTSTEP MTG RR ON SILL SIDE-LH		LH	X	3032.00	1.50	-1.50	0.9	1.4
			Y	-659.60	1.50	-1.50	0.5	-0.7
			Z	532.80	1.50	-1.50	0.2	0.2
FOOTSTEP MTG RR ON SILL SIDE-RH		RH	X	3032.00	1.50	-1.50	-1.2	1.3
			Y	659.60	1.50	-1.50	-1.4	-1.2
			Z	532.80	1.50	-1.50	-0.6	1.4

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
10	Skid mounting bracket location	Confirm the process ,skid mounting bracket is not play any role related to knifing concern . Deliberately skid mounting bracket are damaged ,but no impact observed	Invalid
11	Center floor extension position	Center floor extension bracket damage leads into fouling of rear floor frame assembly at Z08#30 & creates gap in Z axis	**Valid**

TEST OF HYPOTHESIS

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
12	Sill side to rear floor spots	During spot welding with rear floor frame flange get twisted & distortion taken in Z axis. Leads into knifing concern	**Valid**

Flange damage during
spot welding

WHY - WHY Analysis

5.3 Identify the root causes:

Sr No	Probable cause	why	why	Action
1	Z08#10 –Under body fixture Sill side assembly location –Z axis deviation observed 0.1 mm	Additional shim observed in block	No record of shim correction	**Recording of all changes**
2A	2A-Rear floor resting Z22#10 miler fouling to cross member pillar	Less gap between cross member & miler	Less chamfer to miler	Check chamfering of milers
5	Quarter inner panel assembly –nut mismatch	Nut mismatch by 1.8 mm (Not concentric)	Rear Wheel house setting not ok in fixture	**Check setting of parts in fixtures**

WHY - WHY Analysis

5.3 Identify the root causes:

Sr No	Probable cause	why	why	Action
11	Center floor extension position tilted	Fouling in pallet stopper	Pallet stopper position in adequate	**Checks pallet condition**
12	Sill side to rear floor spots twisted	Gun tip position not in @90 deg to part while spot welding	Kick less cable balancing poor	**Check balancing of kick less cable**
7	Rear floor frame assembly –Hole shift	Shifted in piercing operation	No foolproof stopper in piercing	**Check for positive stopper in dies**

6. Implement the corrective measures

6.1 Actions Decided

SN	Root Cause	Action Plan	Target Dt	Resp	Status
1	No record of shim correction	Initiated record for any shim correction for future reference	25.07.12	Vivek	G

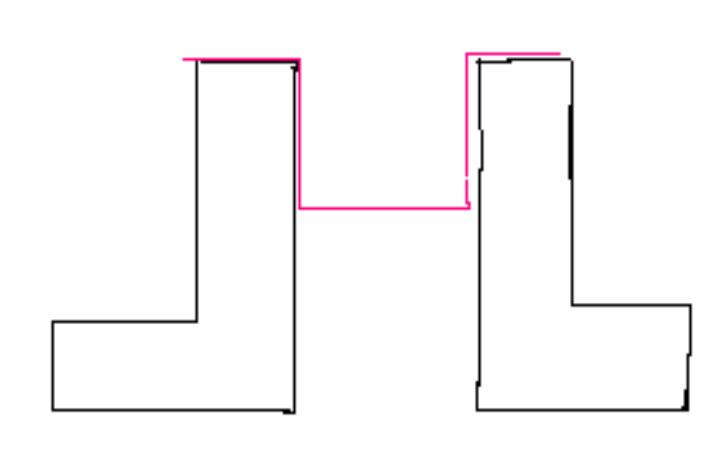
ASSY NAME:	Qtr inner RH				
DATE	UNIT NO	ITEM NO.	CONCERNS	CHANGES DONE/ ACTION TAKEN	REMARKS
10/12/09			Deviation in Upr locating pin	Qtr Inr Upr Loc Pin shifted in +ve X-Direction by 0.5mm	Fixture Calibration.
28/6/10			Inner panel mismatch	Support provided to inner panel	Nova B
10/09/11			Damage at nut plate	Provided nylon guard	I4-379844
12/10/11			Mismatch Trim fixing brkt	Clamp modified by providing stopper	CA3P125
15/4/12			Spot on Jump seat mtg nut	[Provided cu plate guard	I4-85940
18/07/2012			Knifing -C pillar area (Mismatch of 2nd row bottom seat mounting nut	1.0 mm shim correction in Z axis	To eliminate C pillar knifing shim correction done

6. Implement the corrective measures

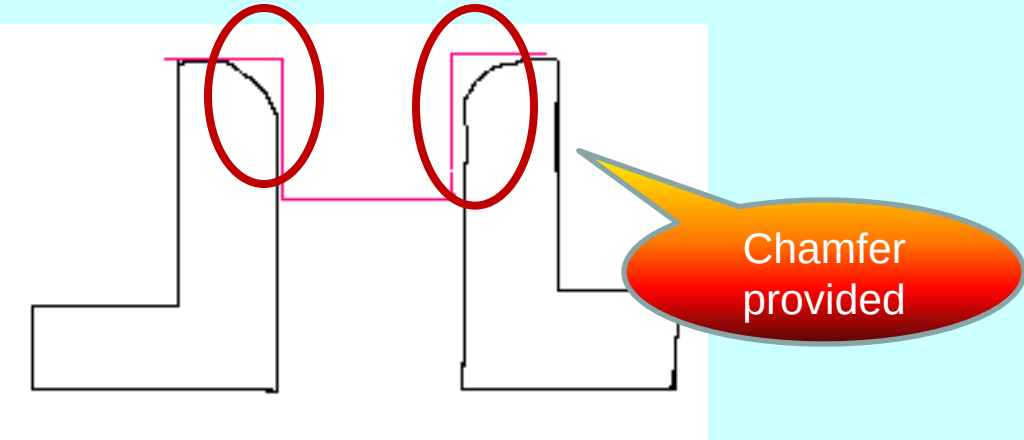
6.1 Actions Decided

SN	Root Cause	Action Plan	Target Dt	Resp	Status
2A	Less chamfer to miler	Provided chamfer to miler (Ensuring after assembly 3.5-4.5 mm between panel)	24.07.12	Vivek	G

BEFORE



AFTER

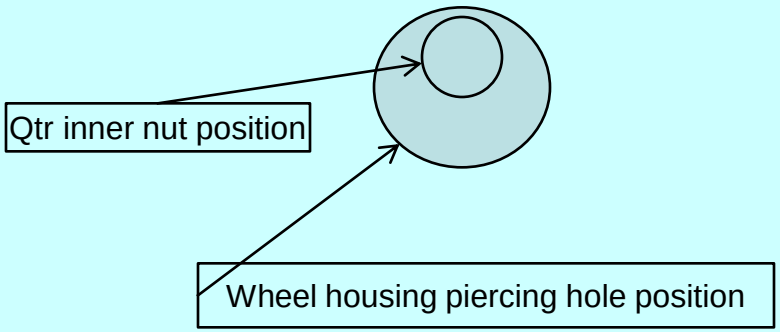


6. Implement the corrective measures

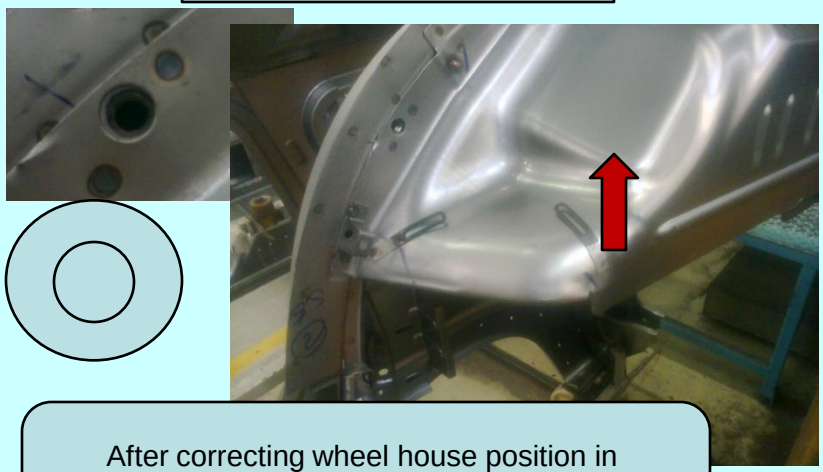
6.1 Actions Decided

SN	Root Cause	Action Plan	Target Dt	Resp	Status
5	Quarter inner panel assembly nut position – Concentricity not ok	Corrected the nut concentricity with respect to rear wheel house.	18.07.12	Vivek	G

BEFORE



AFTER



After correcting wheel house position in Z axis 1.5 mm clearance increased w.r.t.rear floor assembly

6. Implement the corrective measures

6.1 Actions Decided

S N	Root Cause	Action Plan	Target Dt	Resp	Status
11	Center floor pallet-Stopper position in adequate	Corrected all pallets of center floor handling to eliminate damage concern during handling	10.08.12	Prakash	G

BEFORE

AFTER

6. Implement the corrective measures

6.1 Actions Decided

S N	Root Cause	Action Plan	Target Dt	Resp	Status
12	Sill side to rear floor spots twisted	Gun balancing –Kick less cable bunching done as per convenient of operators	25.06.12	Nagar	G

Kick less cable balancing

6. Implement the corrective measures

6.1 Actions Decided

S N	Root Cause	Action Plan	Target Dt	Resp	Status
7	Rear floor frame assembly –Hole shift in Y axis	Provided foolproof stopper in piercing die. In next assembly provided stopper at end, so that defective part can not be seat in assembly fixture	31.08.12	Giri	G

Defective part noticed by Mr. BACHHAV during assembly & prevents Knifing concern

6. Implement the corrective measures

6.1 Actions Decided

POKA-YOKE TO PREVENT THE BRACKET SHIFT

BEFORE	AFTER
Defective part (hole shift) not captured during assy	**Mechanical stoppers provided in both sides of assy fixture to prevent upper cross member shift during assy process**

Problem: Upper cross member of rear floor assy shift

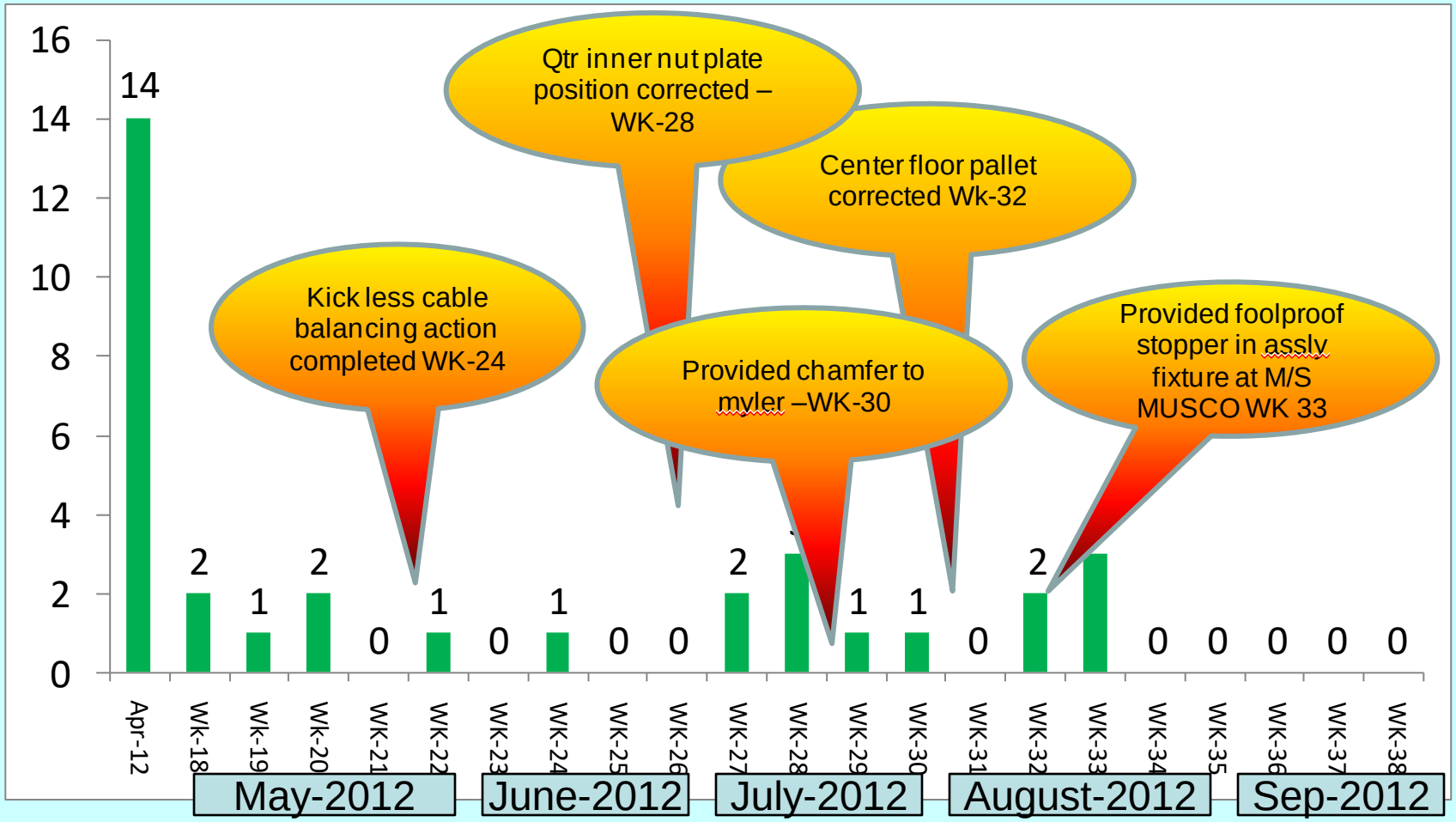
Concern: **Defective part of hole shift not captured during assy of rear floor which causing part shift after assy**

Improvement: **Poka-yoke implemented in Rear Floor Assembly fixture by providing mechanica stoppers to both sides in assy fixture to prevent the upper cross member shift during assy.**

Benefit: Zero customer complaint & internal rejection

7. Check & confirm the results & effects

7.1 Verify Results: No of Defects



7.Check & confirm the results & effects (Cont……)

7.3 Tangible benefits:

Tangible benefits:-

1. 2.21 lacks annualized rework cost at body shop is saved
2. 9 BIW loss per month eliminated-(Cost-4.23 lacs) ,annualized 50.7 lacs is saved
3.Improvement in FTBO from 96.5 to 97.5

Intangible Benefits:

1) Customer satisfaction

2) QC Story approach studied and used for concern resolution

8. Standardisation

8.1 Updation of documents:

1. Fixture history card updated for Z09#20 Rhs

ASSY NAME:	Qtr inner RH				
DATE	UNIT NO	ITEM NO.	CONCERNS	CHANGES DONE/ ACTION TAKEN	REMARKS
3/11/03	F20-03,f10-04,F20-10		Clamp and cylinder hight more to unload the assy due to rotation of Fixture by 180	Clamp and cyl. Removed.	
26/4/04	F20-08,		Wheel house inr seat mtg nut and panel mismatch	2.5 shim added to shift pin Z- direction..5 mm added in X + direction	
26/4/04	F20-10		Wheel house inr seat mtg nut and panel mismatch	1 mm added .	
17/7/06			Butterfly implimentation		EN06-1246D
21/7/06			Butterfly bracket missing	Lifter will not come up unless bracket is welded.	
5/11/06			Butterfly brkt off wlded	crome plated block tried out.	
18/11/06			Butterfly brkt off wlded	modified loc.pin with height & dia provided.	
				Cu block locating provided	
21/05/08			Qtr inner to D plr flange mismatch at side assly #30	2mm shim added in qtr panel assly clamp	
3/7/08			Butterfly bkt offwelded	0.5 mm shim removed in Z direction to take the float to centre.	QCRT L-2
26/10/08			Deviation in Loc Hole On Qtr Inr Upr.	Loc Pin On Qtr Inr Upr shifted in Z-Direction by 0.5mm	
26/10/08			Deviation in Rr Loc Pin On Qtr Lwr.	Loc Pin Rr Loc Pin On Qtr Lwr +ve X-Dir by 0.5mm & +ve Z-Dir by 0.5mm	Fixture Calibration.
12/11/08			Spot pin hole RH butterfly nut plate	Air regulator setting optimized	
10/12/09			Deviation in Upr locating pin	Qtr Inr Upr Loc Pin shifted in +ve X-Direction by 0.5mm	Fixture Calibration.
28/6/10			Inner panel mismatch	Support provided to inner panel	Nova B
10/09/11			Damage at nut plate	Provided nylon guard	I4-379844
12/10/11			Mismatch Trim fixing brkt	Clamp modified by providing stopper	CA3P125
15/4/12			Spot on Jump seat mtg nut	[Provided cu plate guard	I4-85940
18/07/2012			Knifing -C pillar area (Mismatch of 2nd row bottom seat mounting nut	1.0 mm shim correction in Z axis	To eliminate C pillar knifing shim correction done

8. Standardisation

8.1 Updation of documents:

2. Fixture history card updated for Z22#10 Rhs

FIXTURE HISTORY CARD

ZONE: Z22 — **DEPT : MFG ENGG.-SCORPIO BODY**

FIXTURE NO. Z22-10 — **FREQUENCY : 6 Months and/or whenever changes done**

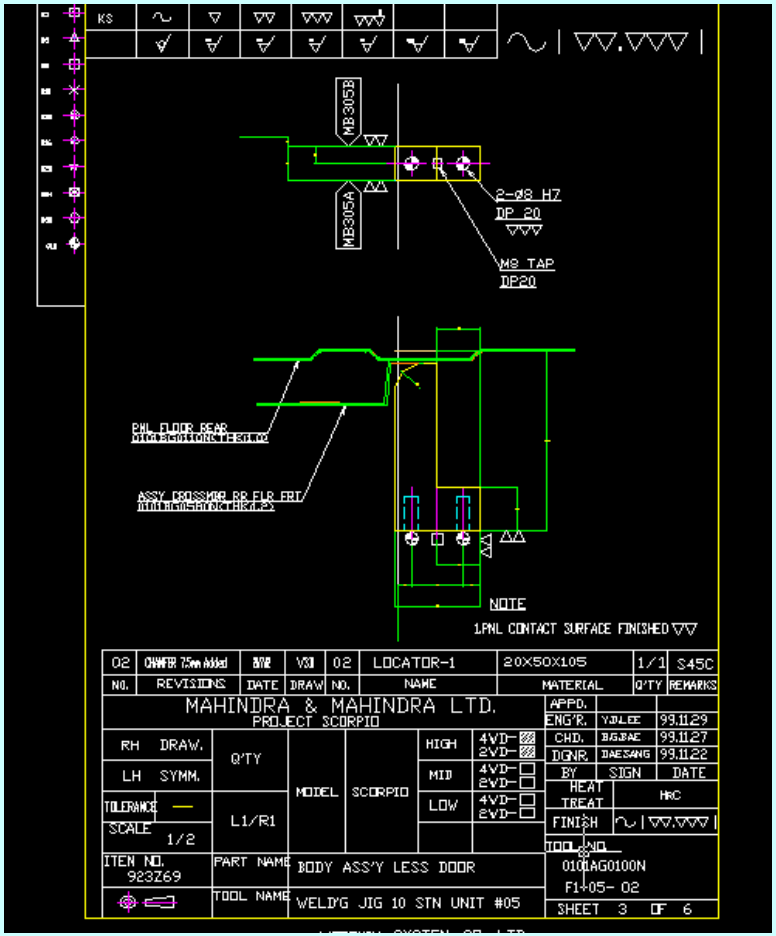
ASSY NAME: Main framing 10

DATE	UNIT NO	ITEM NO.	CONCERNS	CHANGES DONE/ ACTION TAKEN	
16/1/04	F1-14		Damage on Qtr outer near joint of Qtr outer & ring panel.	Nylon block on gate shifted by 3 mm of lh side 2mm of rh side torelease the impression of block on pnl.	
27/1/04	F3-03	F3-03-02,F3-03-03	Side assy to sill side mismatch.side going upside ,reinf.brkt not sitting properly in blocks	Fgap bet'n two blocks increased by 1mm (0.5mm bothsides) to sit the reinf brkt properly.	
23/4/04			Side assy coming out many times after toy tabbing & gates coming out	2.5mm shim removed from the block at A plr lowr on ring panel.	
28/4/04			Roof bow brkt not touching to roof making drumming noise	Shims of block reduced by 3mm ,frt roof bow block removed to reduce gap.	
25/8/06			Damage on Rocker panel	Mettalic resting blocks replaced by Nylon blocks.	
04/06/08			Damage on Rocker panel	Proximaty sensor provide at bottom side rocker panel	
18/3/09			Damage Rocker Panel at Rear Door LH	Guide miler removed.	QCRT CA2P0085/09
20/9/09			Damage qtr outer at lower side	Magnet provided at platform area to restrict hanging side.	QCRT CA3P0037/09
16/06/10			RH Rocker Panel damage near front door area	Nylon guide miler removed	ftbo June-10, NPS
09/30/10			RH Rocker Panel damage near front door area	Nylon guide miler removed	NPS & CAI
01/10/10			Resevoir bkt modification.	Sensor mtg modificattioin done.	EN6436N
17/07/11			Heavy side damage (Lh\Rh)	Proper air pipe brancing done	Scrap July-11
15/09/2011			Damage rocker panel central portion	Side supporting nylon guard modified	FTBO SEPT-11
11/22/11			Damage ring panel Rr Dr opn	Removed guide support	CA2P483
12/15/11			Damage RH Rocker Panel	Rubber insulation provided on B pillar Clamp	CA3P 126
07/01/12			Damage at B Plr area	Rubber roller provided to Balloning side lifting tackle	i4-33512
18/5/12			Damage rocker panel at C plr area	PU material provided as insulation instead of Nylon material for clamp	i4-93243
24/07/2012			To prevent knifing of C pillar at RHS	Provided chamfer to rear floor resting miler to prevent fouling with cross member of rear floor	To eliminate C pillar knifing chamfer provided to miler (Ensuring no fouling of miler with cross member pillar)

8. Standardisation

8.1 Updation of documents:

3. Z22#10 Ballooning fixture miler drawing



Fixture chamfer verified in fixture PM activity as well as while procuring new miler

8. Standardisation

8.1 Updation of documents:

5. Center floor pallet preventive maintenance

	CENTRE FLOOR PALLET		
NAME	**PART NAME**	**OPERATIONS**	
A,B,C,D,E,F,G	VERTICAL PILLAR	W-WELDING	
1,2,3,4,5,6,7,8,9,10	RIGHT LOCKING ASSEMBLY	R-REPLACE	
1',2',3',4',5',6',7',8',9',10'	LEFT LOCKING ASSEMBLY	S-STRAIGHNESS	
L,M,N,O	HOOK FOR LIFTING		
X1,X2,Y1,Y2	FORK ARRESTOR		
	BASE FRAME		
	NUMBER PLATE		
PALLET NO.	**PART TO BE REPAIRED**	**OPRATION**	**DATE**

8. Standardisation

8.2 Training:

Operator were trained to handle TX 158 gun smoothly to prevent spot twist

8.3 Additional Audit / Check

Pallet maintenance activity-Ensuring as per PM

9. Summarise & Plan future actions

9.1 Conclusion:

1.Rear floor frame assembly hole shifting eliminated by providing foolproof stopper in piercing die as well as for detection arrangement made in welding fixture

2.Provided chamfer to resting milers to eliminate fouling with parts

9.2 Horizontal Deployment:

All resting milers check for fouling - 14 milers identified & corrected

9.3 Lessons Learnt:

Things Gone Wrong :Milers matching with resting parts not checked in process.

Things Gone Right : Problem can be solved easily by QC Story approach of the CFT-SYSTEMATIC PROBLEM SOLVING APPROACH.

THANK YOU !!!